

- 1) start scanning from left to right
- 2) If (current element is an operand)
 - Append it to the postfix expression
- 3) Else if (current element is opening bracket '(')
 - Push it onto the stack
- 4) Else if current element is closing bracket ')'
 - Pop elements from the stack and append them to postfix exp till its corresponding opening brackets does not occur
 - Pop opening brackets from the stack and discard both the brackets
- Else
 - // If (current element is an operator)
 - While (stack is not empty && priority of topmost element $\geq$ priority of current element)
 - ✓ Pop element from the stack and append it to postfix expression
 - ✓ Push current element onto the stack.
- 5) Repeat the above steps till end of infix expression
- 6) Pop all the remaining elements from the stack one by one and append them to postfix expression

Infix expression :
 $5+9-4*(8-6/2)+1*(7-3)$

Current element :

Postfix expression: $59+4862/-*-173- \$ +$

stack

top

()
 $\wedge$
 / *
 + -

$5+9-4*(8-6/2)+1*(7-3)$

⑥ ⑦ ⑤ ② ① ⑧ ④ ③
 $5 + 9 - 4 * (8 - 6 / 2) + 1 * (7 - 3)$

$5 + 9 - 4 * (8 - 6 2 /) + 1 * (7 - 3)$

$5 + 9 - 4 * 8 6 2 / - + 1 * 7 3 -$

$5 + 9 - 4 * 8 6 2 / - + 1 7 3 - \$$

$5 + 9 - 4 8 6 2 / - * + 1 7 3 - \$$

$5 9 + - 4 8 6 2 / - * + 1 7 3 - \$$

$5 9 + 4 8 6 2 / - * - + 1 7 3 - \$$

$5 9 + 4 8 6 2 / - * - 1 7 3 - \$ + \rightarrow \text{postfix.}$

59+4862/-*-173-\$+